

Technoeconomic and Life Cycle Analysis of Synthetic Methanol Production from Hydrogen and Industrial Byproduct CO₂

Guiyan Zang,* Pingping Sun, Amgad Elgowainy, and Michael Wang



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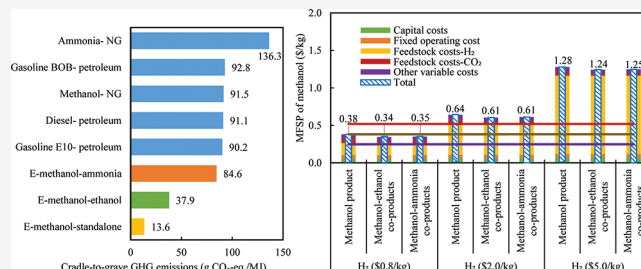


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Supporting Information

ABSTRACT: CO₂ capture and utilization provides an alternative pathway for low-carbon hydrocarbon production. Given the ample supply of high-purity CO₂ emitted from ethanol and ammonia plants, this study conducted technoeconomic analysis and environmental life cycle analysis of several systems: integrated methanol–ethanol coproduction, integrated methanol–ammonia coproduction, and stand-alone methanol production systems, using CO₂ feedstock from ethanol plants, ammonia plants, and general market CO₂ supply. The cradle-to-grave greenhouse gas emissions of methanol produced from the stand-alone methanol, integrated methanol–ethanol, and integrated methanol–ammonia systems are 13.6, 37.9, and 84.6 g CO₂-equiv/MJ, respectively, compared to 91.5 g CO₂-equiv/MJ of conventional methanol produced from natural gas. The minimum fuel selling price (MFSP) of methanol (\$0.61–0.64/kg) is 61–68% higher than the average market methanol price of \$0.38/kg, when using a Department of Energy target renewable hydrogen production price of \$2.0/kg. The methanol price increases to \$1.24–1.28/kg when the hydrogen price is \$5.0/kg. Without CO₂ abatement credits, the H₂ price needs to be within \$0.77–0.95/kg for the MFSP of methanol to equal the average methanol market price. With a CO₂ credit of \$35/MT according to tax credit per metric ton of CO₂ captured and used, the methanol price is reduced to \$0.56–0.59/kg.



INTRODUCTION

The transportation and industry sectors accounted for 36 and 26% of U.S. carbon dioxide (CO₂) emissions from fossil fuel combustion in 2018, respectively, including both direct fossil fuel use and electricity use.¹ Given that CO₂ emission is the main driver for global warming, there has been increased interest in replacing fossil fuels in these sectors by accelerating the deployment of low-carbon fuels.^{2,3} CO₂ capture and utilization (CCU) provides opportunities to synthesize low-carbon hydrocarbon fuels and chemicals by using captured CO₂ as the feedstock and zero-carbon electricity or renewable hydrogen as the energy source.⁴ CCU essentially extends the life of CO₂ over another fuel or chemical product cycle.⁵ Thus, CCU for fuel production can reduce consumption of fossil sources and the resultant emissions.⁶

Although the industrial sector emitted 970 MMT (million metric ton) of CO₂ in 2018,¹ the CO₂ capture from the industrial sector is limited to around 1% of total industrially produced CO₂.⁷ The CO₂ emissions from industrial sources can be distinguished as combustion emissions (occurring from fuel combustion) and process emissions (e.g., due to noncombustion chemical reactions).⁸ Some process emissions from biochemical or chemical reactions, for example, CO₂ emissions from ethanol fermentation and amine separation, have a purity higher than 97%.⁹ With around 44 MMT and 19 MMT CO₂ emissions from ethanol plants (fermentation) and ammonia plants (amine separation) each year, the high-purity

CO₂ supply in the United States is abundant.^{7,10} Given that the higher-purity CO₂ emissions translate into lower capture costs,¹¹ the present study uses high-purity CO₂ from ethanol and ammonia plants as the feedstock for CCU.

Electrofuels (e-fuels) or synthetic hydrocarbons produced from water and waste CO₂ streams, with zero-carbon electricity as the primary energy source, are of increasing interest recently.¹² Among various production routes, synthetic methanol has been extensively researched due to its potential as a low-carbon fuel, energy carrier, or fuel blending component, and the relative ease of its production.¹³ Low- or zero-carbon electricity is used as an energy source to split water and produce hydrogen (H₂) for methanol product in a one-step (directly) or two-step process.¹⁴ The one-step process produces methanol from CO₂ and H₂ directly,¹³ while the two-step process converts CO₂ to CO through a reverse water gas shift (RWGS) reaction and then hydrogenates CO with H₂ to form methanol.¹⁵ Relative to the one-step pathway, the two-step pathway has a higher methanol yield per pass, lower

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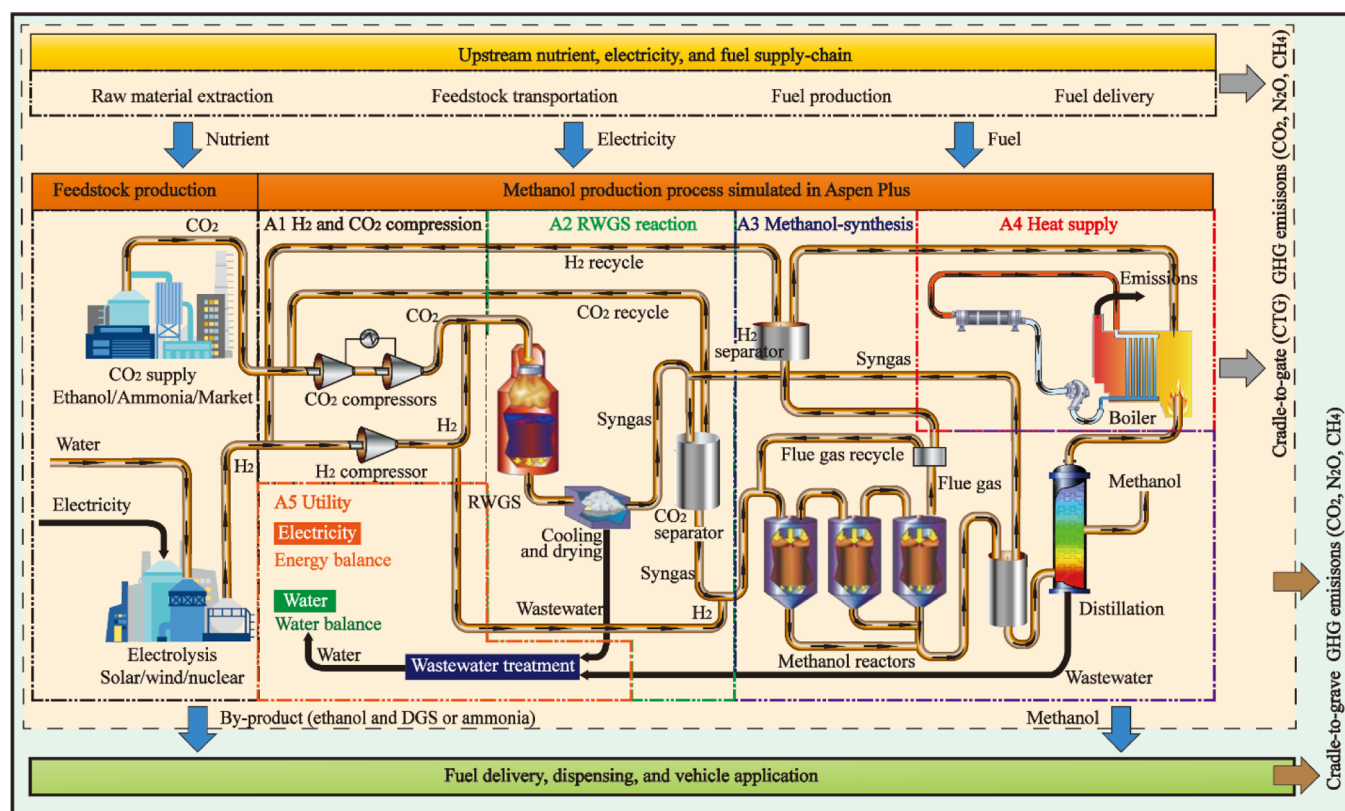


Figure 1. Process flow chart and system boundary of synthetic methanol production for CTG and CTG LCA.

catalyst cost, and smaller hydrogenation reactor size.¹⁶ Thus, this study focuses on the technoeconomic analysis (TEA) and life cycle analysis (LCA) of the two-step pathway of e-fuel methanol synthesis.

Previous TEA analysis results showed that the minimum fuel selling price (MFSP) of synthetic methanol produced from waste CO₂ and H₂ using wind or solar electricity is \$1.00–1.50/kg, more than two times higher than the market methanol price of around \$0.45/kg.^{15,17–20} However, the synthetic methanol pathway can provide significant greenhouse gas (GHG) emission reduction benefit. For example, previous studies showed that synthetic methanol can be produced from waste CO₂ using H₂ from water electrolysis. When the H₂ is produced via electrolysis using renewable electricity, the cradle-to-gate (CTG) GHG emissions of the synthetic methanol are lower than −49.7 g CO₂-equiv/MJ.^{15,17–21} Thus, compared to the CTG GHG emissions of 23.7 g CO₂-equiv/MJ for conventional methanol production from natural gas (does not include methanol combustion CO₂ emissions), the synthetic methanol produced from waste stream CO₂ and renewable electricity is a low-carbon production pathway.²²

Although previous studies evaluated the LCA and/or TEA of e-fuel (synthetic) methanol production in different regions and from various sources, the synthetic methanol production utilizing high-purity CO₂ waste from ethanol and ammonia plants in the United States, considering the expanded system boundary and various hydrogen pathways, has not been thoroughly evaluated. To fill this gap, the LCA and TEA of this study used three CO₂ waste sources from ethanol plants (process CO₂ from fermentation), ammonia plants (with amine scrubbing), and general market CO₂ supply.

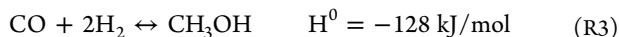
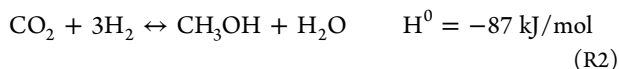
Three CO₂ collecting costs of \$17.3/MT (metric ton), \$20.6/MT, and \$38.6/MT for high-purity CO₂ from the

ethanol plant, the ammonia plant, and market supply were used for the TEA.²³ The LCA of synthetic methanol production was evaluated by defining three system boundaries: integrated methanol–ethanol coproduction system boundary, integrated methanol–ammonia coproduction system boundary, and stand-alone methanol production system boundary. For the LCA and TEA, the synthetic methanol was produced from renewable H₂ (via water electrolysis) and high-purity CO₂ through the two-step reaction process, which was simulated using the Aspen Plus model. The LCA was conducted using the GREET (Greenhouse gases, Regulated Emissions and Energy use in Technologies) 2020 model, whereas the TEA leveraged the H2A model framework.²⁴ Both of the CTG (cradle-to-gate) and CTGR (cradle-to-grave) GHG emissions of the synthetic methanol were evaluated for various H₂ pathways and electricity types. The sensitivity of methanol MFSP to a wide range of H₂ prices and potential CO₂ credits was evaluated.

METHODOLOGY

Synthetic Methanol Production Overview. Using various CO₂ supply options, the present study focused on three systems: methanol–ethanol coproduction, methanol–ammonia coproduction, and stand-alone methanol production, as shown in Figure 1. GREET includes detailed mass, emissions, and energy inventory information for ethanol production (dry milling corn ethanol process with corn oil extraction), ammonia production, and H₂ production. Aspen Plus software was used to simulate the methanol produced from H₂ and CO₂, which provided mass and energy balance and flow information for LCA and TEA. Methanol is produced from H₂ and CO₂ through a two-step pathway: H₂ reacts with

CO₂ to produce CO through an RWGS reaction, denoted as (R1),²⁵ and CO/CO₂ reacts with H₂ to generate methanol through synthesis reactions, denoted as (R2) and (R3).²⁶



RWGS (R1) is endothermic, which requires a high reaction temperature for a reasonable conversion and needs heat supply. Both (R2) and (R3) are exothermic reactions that alleviate the total process of energy demand.

The methanol production process is modeled in five reaction areas (see Figure 1), while the detailed technical parameters are provided in Table S1 (Supporting Information). In area A1, H₂ and CO₂ are compressed to 2.47 MPa.^{27,28} In A2, compressed H₂ and CO₂, with 1:1 M ratio, react in the RWGS reactor at 600 °C and 2.45 MPa to convert 36% of CO₂ into CO.²⁹ A Selexol CO₂ capture unit (A2) is utilized to remove CO₂ and increase the CO molar concentration in the syngas from 31 to 60%.³⁰ The RWGS reaction is based on the experimental result of Kim et al. (2014) using Ba-Ce_{0.2}Zr_{0.6}Y_{0.16}Zn_{0.04}O₃ as the catalyst.²⁹ In reaction area 3 (A3), H₂ and CO gases, with 2.1:1 M ratio, flow through the reactor to synthesize methanol via chemical equilibrium reactions. The methanol reactor has a CO conversion ratio of 17% per pass at 300 °C and 5.1 MPa.³¹ Three reactors in series are used in the Aspen model, with 89% of the effluent gas recycled to the methanol reactors (flue gas recycle in Figure 1), and around 90% of the total CO is converted into methanol. After the effluent gas recycle, 85% of the H₂ is separated by pressure swing adsorption (PSA) and recycled to the H₂ compressor.³² Finally, methanol is distilled and cooled for product storage, while all the combustible components from A3 are fed to the boiler for combustion to provide heat at 900 °C for the RWGS reactor (A4).³³ The last reaction area (A5) accounts for the energy and water balance of the total system, including cooling towers, material storage, and wastewater treatment units. The cooling tower operates at temperatures between 28 and 37 °C with a cooling water recycling efficiency of 99.85%.³⁴

For the stand-alone system, the methanol production efficiency, cold gas efficiency, and the carbon conversion efficiency are defined by eqs 1–3, respectively. For all three system boundaries considered in this analysis, the CTG methanol energy efficiency and CTG total energy efficiency are derived from eqs 4 and 5, respectively.^{35,36}

$$\eta_{E,M} = E_M / (E_{H_2} + W_E) \quad (1)$$

$$\eta_{E,CM} = E_M / (E_{H_2}) \quad (2)$$

$$\eta_{C,M} = (C_M) / C_{CO_2} \quad (3)$$

$$\eta_{E,M,CTG} = E_M / (E_{RE,CTG} + W_{FE,CTG}) \quad (4)$$

$$\eta_{E,T,CTG} = (E_M + E_O) / (E_{RE,CTG} + W_{FE,CTG}) \quad (5)$$

where, $\eta_{E,M}$, $\eta_{E,CM}$, $\eta_{E,M,CTG}$, $\eta_{E,T,CTG}$, and $\eta_{C,M}$ are the methanol production efficiency from H₂ and CO₂, methanol production cold gas efficiency, CTG methanol energy efficiency, CTG total energy efficiency (including energy in coproducts), and

methanol carbon conversion efficiency from CO₂, respectively. E_M is the energy of methanol, E_O is the energy of other coproducts; E_{H_2} and W_E are the H₂ energy and electricity input to the methanol production process from H₂ and CO₂; $E_{RE,CTG}$ is the CTG renewable energy input, and $W_{FE,CTG}$ is the CTG fossil energy input. C_M and C_{CO_2} are the carbon content in methanol and high-purity CO₂ input for the stand-alone methanol produced from H₂ and CO₂.

LCA Methodology. Three systems are defined in this work: stand-alone methanol production system, integrated methanol–ethanol coproduction system, integrated methanol–ammonia coproduction system. The system boundaries defined by inputs and outputs are summarized in Table S4. For the stand-alone system, the CO₂ feedstock is not specified, and the methanol is the only system product (therefore, no allocation is needed). In contrast, for the integrated coproduction system, the CO₂ source is specified, and the methanol process is connected with the CO₂ supply source, for example, corn ethanol plant or ammonia plant. For example, for the integrated methanol–ethanol coproduction system, the system inputs are corn, H₂, electricity, and process fuels, while the system products are methanol, ethanol, dried distillers' grains (DGS), and corn oil. Given the presence of a coproduct other than methanol, the total GHG emissions of the integrated system are allocated to all of the coproducts (methanol, ethanol, DGS, and oil).

Environmental impacts of CTG and CTGR GHG emissions of synthetic methanol production using various boundaries were evaluated using the GREET 2020 model.²² The CTG life cycle accounts for material and energy input for various stages within each boundary including corn growth/transportation (including the nutrients use) and ethanol production process for ethanol plant CO₂ cases, natural gas (NG) recovery and processing and ammonia production process for the ammonia plant CO₂ cases, renewable H₂ production and methanol production process from H₂ and CO₂ of methanol synthesis, and various transportation activities (Figure 1). The CTG life cycle includes all the CTG emissions plus the emissions from methanol's delivery, dispensing, and methanol consumption. The GHG emissions include CO₂, CH₄, and N₂O combined with their global warming potentials of 1, 30, and 265, respectively, using a functional unit of 1 MJ of synthetic methanol.

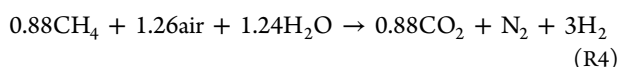
The energy use and GHG emissions of water electrolysis for H₂ production were obtained from GREET 2020, assuming the H₂ transportation distance is 16 km to methanol production sites.³⁷ For the integrated coproduction system boundary (methanol–ethanol or methanol–ammonia), the CO₂ feedstock for methanol production is supplied from the ethanol/ammonia process. A dry milling corn ethanol process with corn oil extraction and an industrial scale ammonia production process (with an amine CO₂ separation) were assumed for this study.^{38,39}

The process-level information for ethanol, ammonia, and methanol production is provided in Table S5 of the Supporting Information. The ethanol and ammonia production utilized mass and energy balance information in GREET, while the methanol production information was obtained from Aspen Plus simulation, as mentioned above.

For the dry milling corn ethanol process, 0.81 kg of DGS and 0.03 kg of oil are coproduced with 1.00 kg of ethanol. The high-purity CO₂ released from the fermentation process is

from industrial data that has a 0.68 of mass ratio (CO_2 /ethanol), calculated from the carbon balance by assuming that corn has a moisture content of 15%⁴⁰ and a dry base carbon content of 45%,⁴¹ while the DGS (dry based) and corn oil have a carbon content of 49%⁴² and 76%,⁴³ respectively. The detailed mass ratio calculation processes are shown in the [Supporting Information](#).

The theoretical conversion processes for ammonia production are listed in reaction (R4) and (R5), with detailed process descriptions shown in the [Supporting Information](#). Ammonia plants have onsite steam methane reforming (SMR) to produce H_2 and CO_2 , which can be separated by either PSA or amine absorption. Currently, approximately 80% of the ammonia plants use amine absorption to produce high-purity CO_2 with a concentration of 97.1 vol %.^{9,44} Thus, the resultant CO_2 is of high purity (~97.1 vol %), which only needs to be compressed for feeding RWGS in the methanol production process. When the high-purity CO_2 is produced from the amine- CO_2 separator, the mass ratio of ammonia to CO_2 is 1:1.09 according to a typical industrial ammonia production process.^{45,46} More ammonia plant information is supplied in the [Supporting Information](#).



For the integrated coproduction system boundary, the energy and emissions burdens are allocated to all system products. For the methanol-ethanol coproduction system, a hybrid allocation method was used in which DGS and oil were treated with a marginal method (i.e., given the displacement credit),⁴⁷ while methanol and ethanol shared process burden through energy [lower heating value (LHV)]. Similarly, the total GHG emissions from the methanol-ammonia coproduction system were allocated to methanol and ammonia based on their energy content, while for the stand-alone methanol production process, all the emissions were assigned to methanol. For the CTG and CTGR base case analysis, H_2 is produced from low-temperature solar/wind electrolysis (with an energy efficiency of 72%), electricity needed for the methanol production from H_2 and CO_2 is supplied by the U.S. mix grid (assuming H_2 and methanol are produced from different locations), and the total emissions of the coproduction systems are allocated to methanol based on energy content. By assuming the LHV of methanol, ethanol, and ammonia are 26.95, 18.90, and 20.09 MJ/kg,⁴⁰ the energy allocation ratios of the methanol in the methanol-ethanol coproduction system and methanol-ammonia system are 25.3 and 41.1%, respectively.

Besides the base case, H_2 produced from LTE nuclear electrolysis (with an energy efficiency of 72%), H_2 produced from high-temperature nuclear electrolysis (with an energy efficiency of 80%), and methanol produced using solar/wind renewable electricity were also studied as alternative cases. [Table S5](#) provides the mass and energy conversion information for ethanol, ammonia, and methanol production based on GREET and Aspen Plus models.

Cost Analysis Methodology. The H2A Production Model Framework was used for the TEA, with 2016 reference year U.S. dollar.²⁴ A discounted cash flow analysis was used to determine the MFSP (MFSP) of methanol to obtain a zero net present value (NPV) in 40 operation years considering total

capital investment (TCI) and operating cost (OC), among other costs.⁴⁸ The key economic and financial assumptions are provided in Table S6 of the [Supporting Information](#).²³

In the TEA model, TCI is calculated according to eq 6 using the total direct capital cost (TDCC) value. TDCC is the sum of the installed equipment costs, scaled from reference equipment cost, as shown in eq 7.²⁴

$$\text{TCI} = \text{TDCC} + 42\% \times \text{TDCC} + \$13.6/\text{m}^2 \times 40,469 \text{ m}^2 \quad (6)$$

$$\text{TDCC} = \sum I_i \times C_{0,i} \times (S_i/S_{0,i})^{f_i} \quad (7)$$

where TDCC is the total direct capital cost, I_i is the installation factor for equipment i ([Table S7](#)), $C_{0,i}$ and $S_{0,i}$ are the equipment cost and equipment size for the reference equipment scale, respectively, S_i is the designed equipment scale from the Aspen Plus model, and f_i is the scaling exponent for each equipment ([Table S7](#)). The total depreciable nonequipment capital cost is $42\% \times \text{TDCC}$, which includes $2\% \times \text{TDCC}$ for site preparation, $10\% \times \text{TDCC}$ for engineering and design, $15\% \times \text{TDCC}$ for project contingency, and $15\% \times \text{TDCC}$ for upfront permitting costs. The estimated land footprint cost is $\$13.6/\text{m}^2 \times 40,469 \text{ m}^2$.

The OC includes fixed OC (FOC) and variable OC (VOC) using the calculation processes shown in eqs 8 and 9.

$$\text{FOC} = \text{LC} + 20\% \times \text{LC} + 2\% \times \text{TCI} + 0.3\% \times \text{TCI} \quad (8)$$

$$\text{VOC} = \sum \dot{m}_i \times C_i \quad (9)$$

In eq 8, LC is the burdened labor cost, including overhead for a labor number of 68, $20\% \times \text{LC}$ is the general and administrative expense, $2\% \times \text{TCI}$ is for the property tax and insurance, and $0.3\% \times \text{TCI}$ is the annual maintenance and repair cost. In eq 9, $\dot{m}_i$ is the annual mass consumption of H_2 and CO_2 feedstock, industrial electricity consumption, cooling water use, process water use, and catalyst that are supplied by the Aspen Plus model, while C_i is the price of the individual inputs and materials.

For the TEA, various H_2 prices were discussed from H2A default cases, and $\$2.0/\text{kg}$ for H_2 , the Department of Energy (DOE) Hydrogen and Fuel Cell Technologies Office (HFTO) target price produced from water central electrolysis, is selected as a reference for the base case study.⁴⁹ For the three different systems of methanol-ethanol coproduction, methanol-ammonia coproduction, and methanol stand-alone production, the CO_2 costs are assumed to be $\$17.3/\text{MT}$, $\$20.6/\text{MT}$,⁹ and $\$38.6/\text{MT}$,⁵⁰ respectively. The CO_2 prices of $\$17.3/\text{MT}$ and $\$20.6/\text{MT}$ accounts for the cost of separation, purification, and compression,⁹ while $\$38.6/\text{MT}$ represents the traded CO_2 market price contracted by a Dakota Gasification Company.⁵⁰ The prices of process water, cooling water, and industrial electricity are assumed to be $\$0.63/\text{MT}$, $\$0.03/\text{MT}$, and $\$0.07/\text{kWh}$, respectively.²⁴ The total catalyst cost for RWGS and methanol synthesis is $\$2254\text{k}$ (k is thousand), replaced once every three years.⁵¹

RESULTS

Process Modeling Results. The modeled process produces 1190 MT/day methanol by consuming 243 MT/day H_2 and 1978 MT/day CO_2 (with a molar ratio of 2.7:1.0) ([Table S2](#) and [Figure S1](#) in the [Supporting Information](#)).

Table 1. Performance Analysis Results of the Three Systems

	methanol production system	methanol–ethanol coproduction system	methanol–ammonia coproduction system
H ₂ consumption (MT/day) ^a	243	243	243
high-purity CO ₂ consumption (MT/day) ^a	1978	0	0
ethanol production (MT/day) ^b	0	2912	0
ammonia production (MT/day) ^c	0	0	1817
DGS production (MT/day) ^b	0	2,369	0
corn oil production (MT/day) ^b	0	83	0
methanol production (MT/day) ^a	1190	1190	1190
total carbon conversion efficiency (%) ^a	82.5%	82.5%	82.5%
H ₂ input for methanol synthesis (MW) ^a	337.8	337.8	337.8
electric power input for methanol synthesis (MW) ^a	28.0	28.0	28.0
CTG renewable input (MW) ^d	485.1	2327.6	489.8
CTG fossil input (MW) ^d	49.4	514.9	905.4
ethanol (MW) ^b	0.0	908.4	0.0
ammonia (MW) ^c	0.0	0.0	397.4
DGS (MW) ^b	0.0	555.1	0.0
corn oil (MW) ^b	0.0	35.8	0.0
methanol (MW) ^d	276.7	276.7	276.7
methanol synthesis energy efficiency (%) ^a	75.6%	75.6%	75.6%
CTG methanol production efficiency (%) ^d	51.8%	9.7%	19.8%
CTG total energy efficiency (%) ^d	51.8%	62.5%	48.3%

^aData are from the Aspen model of methanol produced from H₂ and CO₂. ^bData are from the methanol–ethanol coproduction system. ^cData are from the methanol–ammonia coproduction system. ^dData are from the CTG energy analysis including upstream fuel and nutrient supply chain.

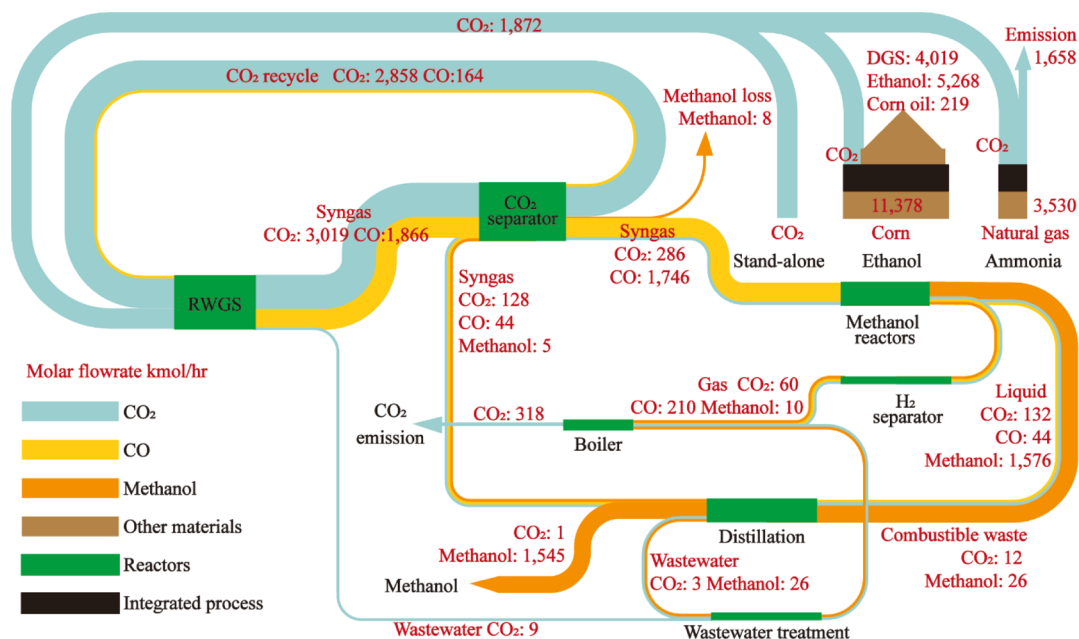


Figure 2. Sankey diagram of the carbon balance of three systems. All the values are in the unit of kmol/h. The width of flow expresses the quantity of carbon molar flow rate in different components, the green and black boxes represent conversion processes and integrated process, respectively, and the blue-yellow, orange, brown flows represent CO₂, CO, methanol, and other material flows, respectively.

When this CO₂ is supplied by the ethanol plant or ammonia plant, the integrated system produces 2912 MT/day of ethanol, 2369 MT/day of DGS, and 83 MT/day of corn oil, or 1817 MT/day of ammonia (Table 1).

Table 1 shows that the methanol production efficiency is 75.6%, considering the total process energy input, that is, including both H₂ and electricity. Whereas for the stand-alone methanol production system, the CTG methanol production efficiency is 51.8%, considering the CTG total energy input, that is, including the upstream energy use for H₂ production.

When the methanol production process is integrated with the ethanol or ammonia production process, the methanol CTG energy efficiency is 62.5 and 48.3%, respectively. The detailed material consumptions and CTG energy consumption for methanol, ethanol, and ammonia stand-alone production processes are shown in Tables S3 and S5.

In the methanol production process, 82.5% of carbon in CO₂ feedstock is converted into methanol with the detailed carbon balance shown in Figure 2. In Figure 2, for the stand-alone methanol system, 82.5% of carbon from CO₂ feedstock

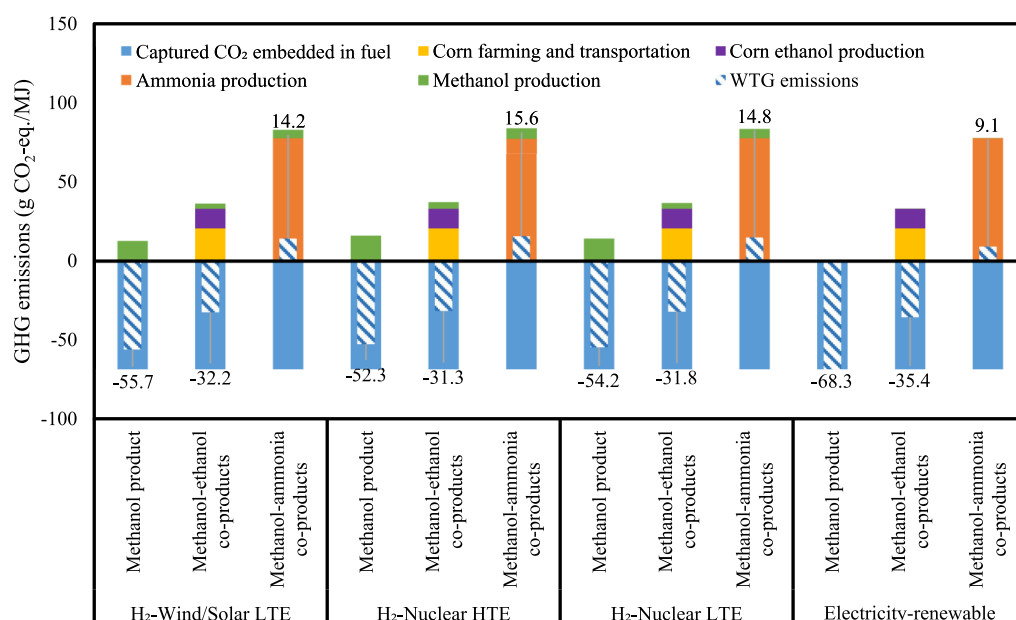


Figure 3. Methanol CTG (cradle-to-gate) GHG emissions with various H₂ sources, electricity types, and allocation methods. Methanol is the single product from the methanol production system, while emissions from the (integrated) coproduction systems are allocated to ethanol and methanol or ammonia and methanol. The base cases use wind/solar H₂, U.S. average grid mix electricity, and energy allocation (LHV).

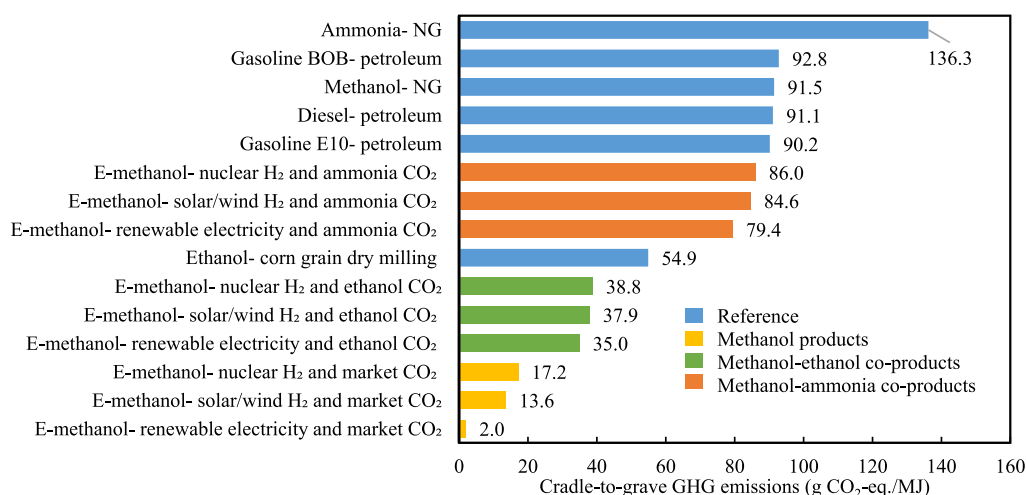


Figure 4. CTGR (cradle-to-grave) GHG emissions of synthetic methanol compared with other chemicals and fuels. The blue bars are GHG emissions of conventional chemicals and petroleum fuel pathways from GREET, the yellow bars are the GHG emissions of synthetic methanol from the stand-alone production system, while the green and orange bars are the GHG emissions of methanol from the methanol–ethanol and methanol–ammonia coproduction systems, respectively. BOB is short for blend stock for oxygenate blending. The cases use wind/solar LTE for H₂ production and the U.S. average electricity grid mix for the methanol synthesis process and energy allocation if the details are not otherwise mentioned.

(1872 kmol/h) is converted into methanol (1545 kmol/h), with 17.0% carbon emitted from combustion of tail gas in the boiler, and the other 0.5% carbon emitted from CO₂ capture unit. For the methanol–ethanol coproduction system, the carbon in corn is the carbon source for the total system; in the ethanol production process, 46.3, 35.3, and 1.9% of carbon in corn have been converted into ethanol, DGS, and oil, respectively, while the 16.5% remaining carbon is released as high-purity CO₂, which is used for methanol production. The total carbon conversion ratio of the methanol–ethanol coproduction system is 97.1%, that is, 13.6% higher than that of the corn ethanol (stand-alone) plant. For the methanol–ammonia coproduction system, NG is the carbon

source. In the ammonia production process, 53.0% of the NG carbon is produced as high-purity CO₂, which is used for methanol production, resulting in a total carbon conversion ratio of 43.8% for the methanol–ammonia coproduction system.

The result of methanol synthesis modeling is also compared with previous studies. For example, the work of Anicic et al. showed the two-step methanol synthesis process has a cold gas efficiency of 80.1% and a carbon conversion ratio of 79.0%.¹⁴ It is worth noting that the cold gas efficiency is defined as the ratio of the energy in the methanol product divided by energy in H₂ feedstock without accounting for the electricity consumption (shown in eq 2). To allow comparison on a

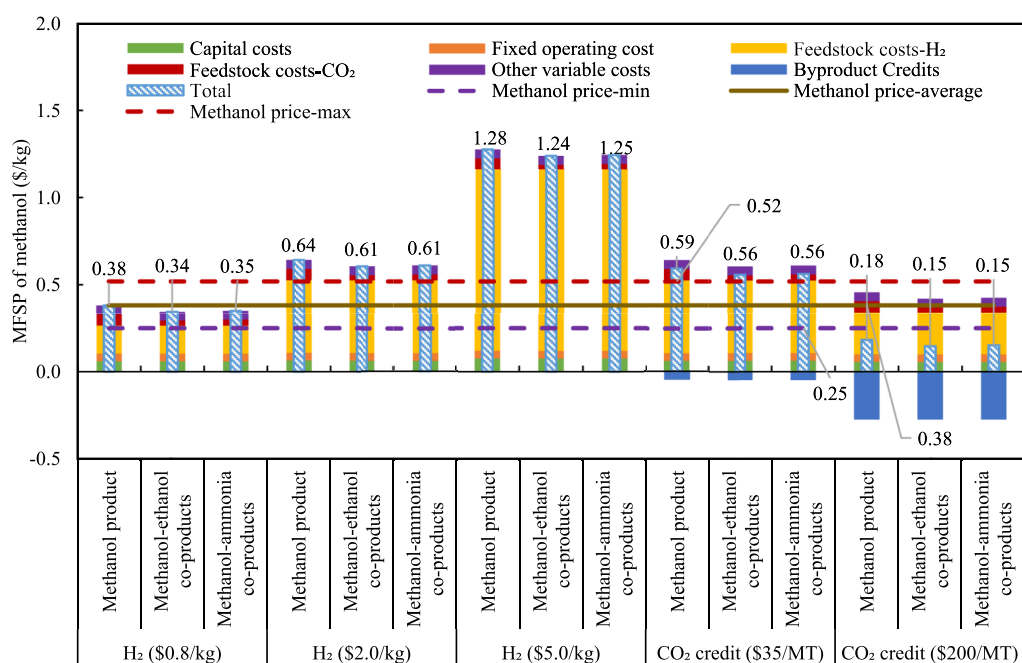


Figure 5. Variation of MFSP with H₂ price and potential CO₂ credit.

consistent basis, the present study also calculated the cold gas energy efficiency, without accounting for electricity consumption, leading to a higher efficiency of 81.9% and a higher carbon conversion ratio of 82.5%. The small differences in cold gas efficiency and carbon conversion ratio between this study and Anicic et al. are attributed to the different design parameters for the RWGS and methanol synthesis processes, such as the reaction temperature, pressure, conversion ratio, and hydrogen recycle ratio.

LCA Results. The CTG GHG emissions of the three considered methanol production systems were estimated with various H₂ production and electricity generation technologies and coproduct allocation methods, as shown in Figure 3. The base case study uses H₂ production from LTE of water using wind/solar power and energy allocation (with marginal method approach for DGS and oil from the ethanol–methanol coproduction system). The CTG GHG emissions (without methanol combustion CO₂ emissions) of the stand-alone methanol production in the base case is −55.7 g CO₂-equiv/MJ methanol with −68.4 g CO₂-equiv/MJ from captured CO₂ embedded in methanol (blue bars in Figure 3) and 12.7 g CO₂-equiv/MJ emitted, mostly from the upstream grid electricity used for the methanol synthesis process.²² When the methanol production process is integrated with the ethanol plant or ammonia plant, the total system burdens are shared or allocated by energy to both methanol and ethanol/ammonia, thus mitigating the burden on ethanol/ammonia while increasing the methanol CTG GHG emissions to −32.2 g CO₂-equiv/MJ and 14.2 g CO₂-equiv/MJ for the methanol–ethanol and methanol–ammonia coproduction systems, respectively.

In addition to the H₂ produced from wind/solar LTE, two nuclear energy base H₂ production pathways of HTE and LTE were also studied. In general, the GHG emissions from nuclear HTE–H₂ pathways are 0.5–1.9 g CO₂-equiv/MJ lower compared to LTE–H₂ pathways but are 0.4–1.5 g CO₂-equiv/MJ higher compared to the wind/solar–H₂ pathway. When the U.S. average grid mix electricity is replaced by

renewable electricity, the CTG GHG emissions of the stand-alone methanol production system are reduced to −68.3 g CO₂-equiv/MJ, and that of the methanol–ethanol and methanol–ammonia co-production systems decrease by 3.2 g CO₂-equiv/MJ and 5.1 g CO₂-equiv/MJ, respectively.

The CTGR GHG emissions of synthetic methanol from the three aforementioned production systems are compared with the emissions from other fuels, as shown in Figure 4. Methanol from the stand-alone system with renewable electricity supply has the lowest CTGR GHG emission of 2.0 g CO₂-equiv/MJ as compared to 91.5 g CO₂-equiv/MJ for the conventional methanol production from NG (i.e., 89.5 g CO₂-equiv/MJ CTG GHG emission reduction). Using U.S. average grid electricity for the methanol synthesis process, the stand-alone system has the potential to reduce CTGR GHG emissions by 74.3–77.9 g CO₂-equiv/MJ with various renewable and nuclear hydrogen sources, compared to the NG–methanol pathway. When the methanol production process is integrated with the corn ethanol plant (methanol–ethanol coproduction system), the CTGR GHG emissions of methanol are 52.7–56.5 g CO₂-equiv/MJ lower compared to the NG–methanol pathway. The CTGR GHG emissions of methanol, considering the integrated methanol–ammonia coproduction system with various electricity sources and hydrogen sources, are 5.5–12.1 g CO₂-equiv/MJ lower compared to the NG–methanol pathway.

Economic Analysis Results. In this study, the CO₂ prices are assumed to be \$17.3/MT, \$20.6/MT, and \$38.6/MT for the methanol–ethanol coproduction system, methanol–ammonia coproduction system, and stand-alone methanol production system, respectively. The TCI of the stand-alone methanol production is \$305 million (MM), with a TDCC of \$213 MM (details are provided in Tables S7 and S8 of the Supporting Information). For the methanol production process, the RWGS reaction area, methanol-synthesis area, and utility area have the largest installed equipment cost of \$76 MM, \$50 MM, and \$40 MM that are 35.8, 23.3, and 19.0% of the total depreciable capital cost, respectively. The total

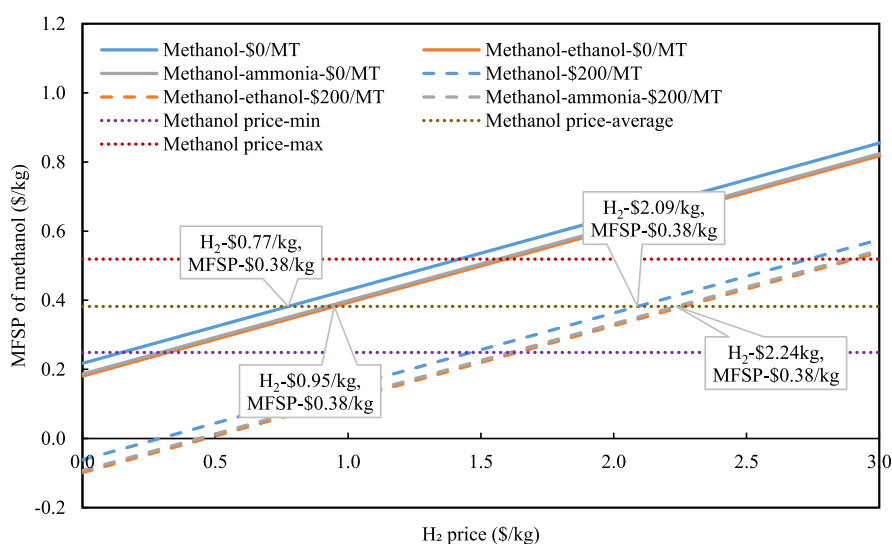


Figure 6. Breakeven H_2 prices for different methanol production systems and various CO_2 credit values. In the legend, \$0/MT refers to a CO_2 credit of \$0/MT, and \$200/MT refers to CO_2 credit of \$200/MT. The call-out shows the breakeven H_2 price such that the MFSP of methanol equals the average methanol market price of \$0.38/kg.

variable OC is \$190 MM/year when the H_2 price is \$2.0/kg and CO_2 price is \$17.3/MT (Table S9). While the total fixed OC is \$17 MM/year, with a labor number of 68 and a burdened labor cost including overhead of \$58/man h (Table S8).

Figure 5 shows the MFSP of synthetic methanol as compared to the average methanol market price over the past five years of \$0.38/kg. When the DOE target hydrogen production price of \$2.0/kg is assumed, the MFSP of the methanol from the stand-alone system is \$0.64/kg, and from the coproduction systems (methanol–ethanol and methanol–ammonia) is \$0.61/kg. The evaluated MFSPs are 17–23% higher than the maximum methanol price in the past five years and 61–68, 144–156% higher than the average and minimum methanol prices in the past five years. As shown in Figure 5 for the methanol–ethanol coproduction system, feedstock costs for H_2 (\$2.0/kg) and CO_2 (\$17.3/MT) and the capital costs dominate the methanol MFSP, contributing 68.9, 4.8, and 10.5%, respectively. When H_2 is produced from renewable electricity using the current renewable polymer electrolyte membrane electrolysis (PEME) technology, the H_2 price increases to \$5.0/kg,⁴⁹ resulting in a higher MFSP of methanol (\$1.24–1.28/kg). However, given that the low CTGR GHG emission of the stand-alone methanol production system is 2.0–17.2 g CO_2 -equiv/MJ, the synthetic methanol production using renewable hydrogen can be qualified as a low-carbon fuel compared to conventional methanol production from NG (with a CTGR GHG emissions of 91.5 g CO_2 -equiv/MJ). If avoided CO_2 can be traded in a carbon trading market, the potential carbon credit can further reduce the MFSP of synthetic methanol, as shown in Figure 5. According to the Section 45Q carbon capture tax credit, the CO_2 utilization for synthetic methanol production has a CO_2 credit of \$35/MT.⁵² With a CO_2 credit of \$35/MT, the synthetic methanol MFSP is \$0.56–0.59/kg, which is \$0.05/MT lower than the cases with zero CO_2 credit. If the CO_2 credit increases to \$200/MT,⁵³ the synthetic methanol MFSP from the stand-alone production system is \$0.18/kg and from the methanol–ethanol and methanol–ammonia coproduction systems is \$0.15/kg, which are 52.6 and 60.5% lower than the average

methanol market price in the past five years. Given that the variation of the TEA assumptions results in uncertainty of the MFSP, MFSP cost distributions, sensitivity analysis, and uncertainty analysis results are supplied in the Supporting Information Figures S4–S6.

Figure 6 shows the breakeven H_2 prices for the MFSP of methanol to equal the average methanol market price (\$0.38/kg) for different methanol production system boundaries and various CO_2 credit values. For the cases without CO_2 credit, the breakeven H_2 prices for the three evaluated systems are within \$0.77–0.95/kg. However, for the cases with \$200/MT CO_2 credit, the breakeven H_2 prices are \$2.09–2.24/kg. The TEA results indicate that the economics of synthetic methanol production can be improved by developing a low-cost water electrolysis technology for renewable H_2 production and leveraging credits in carbon trading markets.

■ IMPLICATIONS

The present study evaluated TEA and LCA of synthetic methanol using high-purity CO_2 from ethanol and ammonia plants, with various price scenarios. Simulation results show that integrating methanol synthesis with ethanol and ammonia production improves the production energy efficiency, carbon conversion efficiency, and GHG emissions.

This study used high-purity CO_2 emissions from typical ethanol and ammonia plants as feedstock for the synthetic methanol production. The integrated system boundary treatment can be extended to other industries such as cement, iron and steel, carbonates, and petrochemicals. Moreover, if the current results are scaled to combine with the industrial distribution data of the ethanol and ammonia plants in the U.S., the potential total H_2 market demand and the potential total synthetic methanol production, as well as their environmental impacts can be further evaluated.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.0c08237>.

Additional details on simulation assumptions, flow results, mass and energy balance of methanol production from H₂ and CO₂; system boundary of stand-alone and integrated systems; ethanol production processes; ammonia production processes; TEA assumptions and cost results; methanol MFSP components; and sensitivity analysis and risk analysis of MFSP of methanol (PDF)

AUTHOR INFORMATION

Corresponding Author

Guiyan Zang — Systems Assessment Center, Energy Systems Division, Argonne National Laboratory, Lemont, Illinois 60439, United States; orcid.org/0000-0003-0948-4673; Email: gzung@anl.gov

Authors

Pingping Sun — Systems Assessment Center, Energy Systems Division, Argonne National Laboratory, Lemont, Illinois 60439, United States; orcid.org/0000-0001-8226-9270

Amgad Elgowainy — Systems Assessment Center, Energy Systems Division, Argonne National Laboratory, Lemont, Illinois 60439, United States

Michael Wang — Systems Assessment Center, Energy Systems Division, Argonne National Laboratory, Lemont, Illinois 60439, United States

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.est.0c08237>

Notes

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ABBREVIATIONS

BOB	blend stock for oxygenate blending
CO ₂	carbon dioxide
CCU	carbon capture and utilization
DCFROR	discounted cash flow rate of return
DOE	Department of Energy
E-fuels	electrofuels
FOC	fixed operating costs
GHG	greenhouse gas
REET	greenhouse gases, regulated emissions, and energy use in transportation
H ₂	hydrogen
HFTO	Hydrogen and Fuel Cell Technologies Office
LCA	life cycle analysis
LHV	lower heating value
LTE	low-temperature electrolysis
MFSP	minimum fuel selling price
NG	natural gas
NPV	net present value
OC	operating cost
PEME	polymer electrolyte membrane electrolysis
PSA	pressure swing adsorption

RWGS	reverse water gas shift
SMR	steam methane reforming
SOEC	solid oxide electrolyzer cell
TCI	total capital investment
TDCC	total direct capital cost
TEA	technoeconomic analysis
HTE	High-temperature electrolysis
VOC	variable operating costs
CTG	cradle-to-gate
CTGR	cradle-to-grave

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